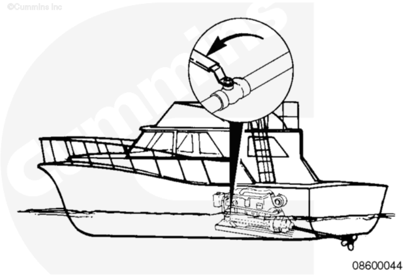
Trainings

Flush

⚠ CAUTION ⚠

Zinc plugs expand and can break off during removal. Inspect the zinc plug to make sure it is in one piece. Refer to Procedure 008-059 in Section 8 (/qs3/pubsys2/xml/en/procedures/40/40-008-059.html). If the plug is broken, it must be replaced with a new zinc plug, and the broken pieces must be retrieved from the aftercooler to prevent damage to components downstream in the sea water system.

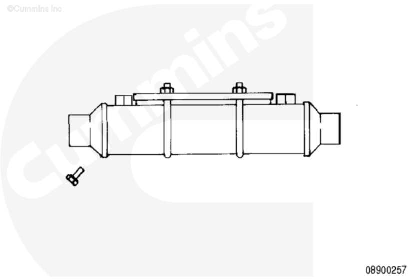


Depending on the engine option selected, the gear oil cooler could be mounted on the rear or side of the engine.

Shut OFF the sea water inlet valve on the vessel hull, if equipped.

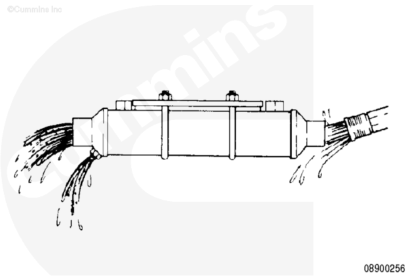
Remove the marine gear oil cooler drain plug/zinc annode. Drain the sea water from the gear oil cooler.

Disconnect the sea water inlet and outlet connections.



Use clean water to back flush all the debris from the cooler.

Make sure the debris flushed from the cooler does **not** enter the water supply hoses.

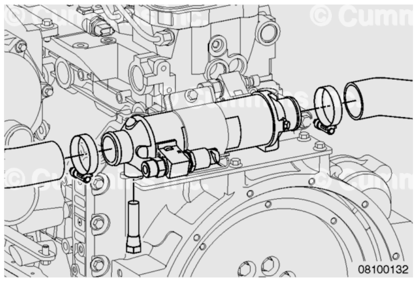


Rear Mount Gear Oil Cooler:

Install the drain plug/zinc anode and sea water hose connections.

Inspect and install the zinc plugs in the gear oil cooler. Refer to Procedure 008-059 in Section 8. (/qs3/pubsys2/xml/en/procedures/40/40-008-059.html)

Note : Do not use thread sealant on the zinc plugs. They must be grounded to the component to function properly.

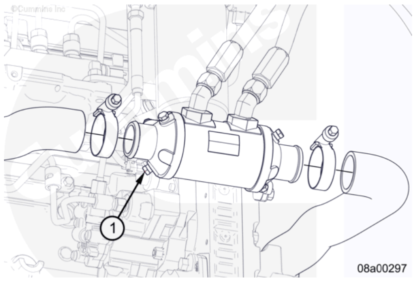


Side Mount Gear Oil Cooler:

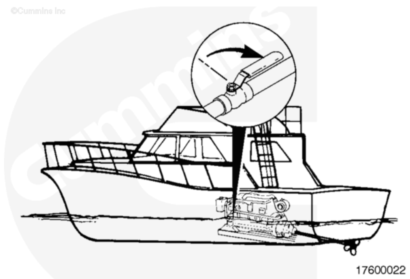
Install the drain plug/zinc anode (1) and sea water hose connections.

Inspect and install the zinc plugs in the gear oil cooler. Refer to Procedure 008-059 in Section 8. (/qs3/pubsys2/xml/en/procedures/40/40-008-059.html)

Note : Do not use thread sealant on the zinc plugs. They must be grounded to the component to function properly.



Open the sea water valve on the vessel hull, if closed.



Preparatory Steps

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

⚠ CAUTION ⚠

Use caution when disconnecting or removing oil lines that oil is not spilled or drained into the bilge area. The oil must be drained into a suitable container and disposed of in accordance with local environmental regulations.

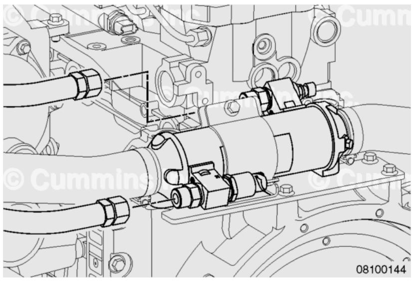
- Close the sea water inlet valve(s) on the vessel hull, if equipped. See equipment manufacturer service information.
- Disconnect the battery power to the engine. See equipment manufacturer service information.
- Drain the sea water system. Refer to Procedure 010-005 in Section 10. (/qs3/pubsys2/xml/en/procedures/40/40-010-005-tr.html)
- Remove the air cleaner element.

Remove

Rear Mount Gear Oil Cooler:

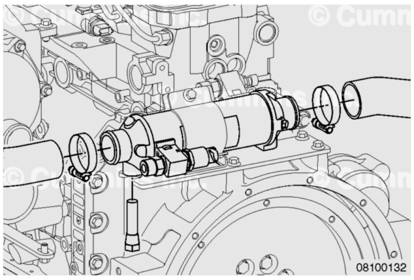
Remove the transmission oil supply and return lines. Mark their locations.

Disconnect the marine gear oil pressure and temperature sensor, if equipped.



⚠ CAUTION ⚠

Removing the sea water supply line for the shaft log seal will allow sea water to leak into the vessel, if the line is not plugged. To reduce the possibility of damage, be sure to plug the supply line and prevent sea water from leaking into the vessel.



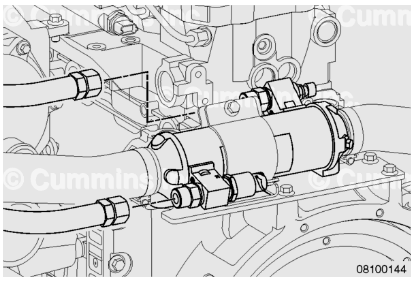
Remove the aftercooler sea water discharge hose from the gear cooler.

Remove the marine gear cooler sea water discharge hose from the gear cooler.

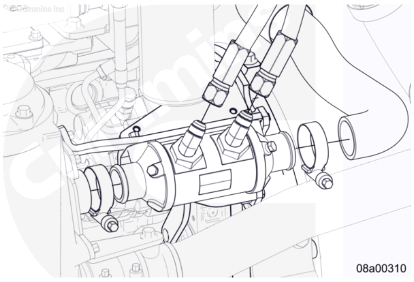
Remove and plug the sea water supply line for the shaft log seal from the gear oil cooler, if equipped.

Remove the bracket-to-cooler mounting capscrews.

Remove the cooler bracket-to-flywheel housing mounting capscrews and marine gear oil cooler.



Side Mount Gear Oil Cooler:



⚠ CAUTION ⚠

Removing the sea water supply line for the shaft log seal will allow sea water to leak into the vessel, if the line is not plugged. To reduce the possibility of damage, be sure to plug the supply line and prevent sea water from leaking into the vessel.

Remove the aftercooler sea water discharge hose from the gear cooler.

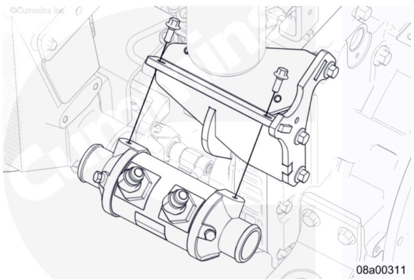
Remove the marine gear cooler sea water discharge hose form the gear cooler.

Remove and plug the sea water supply line for the shaft log seal from the gear oil cooler, if equipped.

Remove the transmission oil supply and return lines. Mark their locations.

Disconnect the marine gear oil pressure/temperature sensor, if equipped.

Remove the marine gear oil cooler by removing the bracket mounting capscrews.



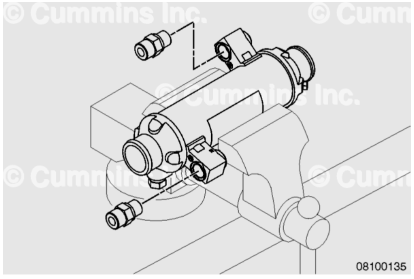
If necessary for repair of gear oil cooler bracket, remove the aftercooler bracket mounting capscrews.

If necessary for repair of gear oil cooler bracket, remove the flywheel housing bracket mounting capscrews.

Disassemble

Rear Mount Gear Oil Cooler:

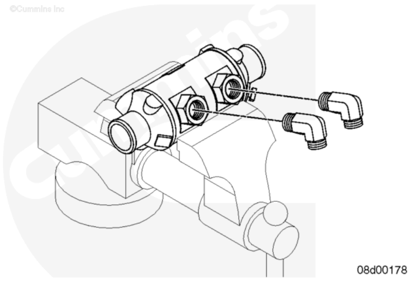
Place the cooler in a vise. Remove the two oil line fittings from the gear cooler, if necessary. Mark the fitting locations and orientation prior to removal.



Side Mount Gear Oil Cooler:

Mark the fitting locations and orientation prior to removal.

Place the cooler in a vise. Remove the two oil line fittings from the gear cooler, if necessary.

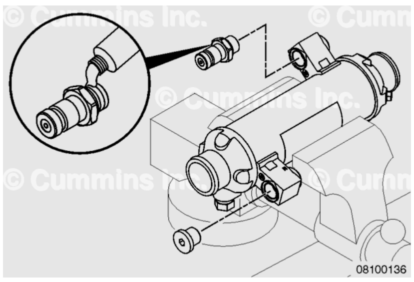


Inspect for Reuse

Rear Mount Gear Oil Cooler

Plug one gear oil port and attach an air supply line to the other gear oil port with a quick disconnect fitting. Apply thread sealant to the threads to prevent leaks. Do **not** allow sealant to enter the gear oil cooler.

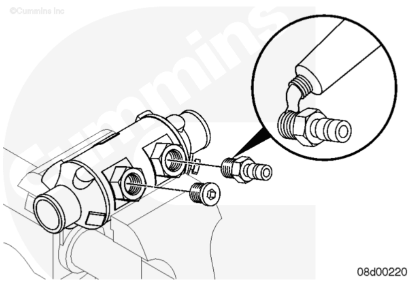
Note : If available, use a plug and quick disconnect fitting that seals with an o-ring fitting. A thread sealant will not need to be used in this instance.



Side Mount Gear Oil Cooler

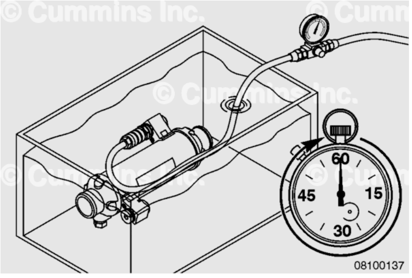
Plug one gear oil port and attach an air supply line to the other gear oil port with a quick disconnect fitting. Apply thread sealant to the threads to prevent leaks. Do **not** allow sealant to enter the gear oil cooler.

Note : If available, use a plug and quick disconnect fitting that seals with an o-ring fitting. A thread sealant will not need to be used in this instance.



⚠ WARNING ⚠

Troubleshooting with high-pressure air presents the risk of equipment damage, personal injury, or death. Troubleshooting must be performed by trained, experienced technicians.



⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

Note : The illustrations for pressure testing show the rear mount gear oil cooler; the steps for the side mount gear oil cooler are the same.

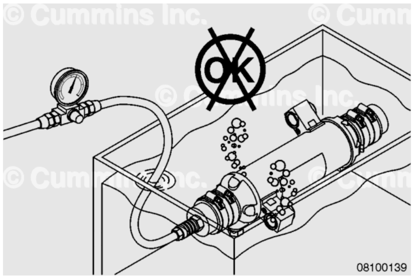
Attach a high-pressure air supply source (air cylinder or other suitable source) with an air pressure regulator and an inline shutoff valve to the quick disconnect fitting.

Set the regulator test pressure to:

- Rear Mount Gear Oil Cooler - 689 kPa [100 psi]
- Side Mount Gear Oil Cooler - 1724 kPa [250 psi].

Submerge the gear oil cooler into a tank of water. Rotate the cooler to allow any trapped air to escape. Allow the cooler to remain submerged for 1 minute.

Inspect for air bubbles at the fitting braze joints.



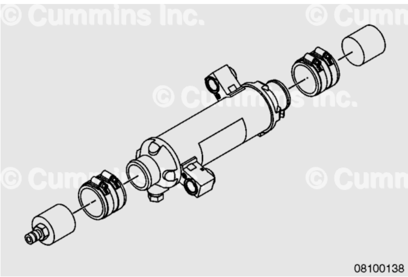
Inspect for air bubbles at the opening at each end of the cooler.

If leaks are detected, replace the gear oil cooler.

Remove the cooler from the tank. Shut off the air supply and disconnect the air supply.

Remove the plug and test fitting.

Fabricate a test fixture to seal the sea water connections, or use connector hoses with a quick disconnect air connection to supply a regulated test pressure of 276 kPa [40 psi] to the sea water side of the gear oil cooler.



Note : Although the rear mount cooler is shown, this applies to both cooler designs.

Submerge the cooler into a tank of water for one minute.

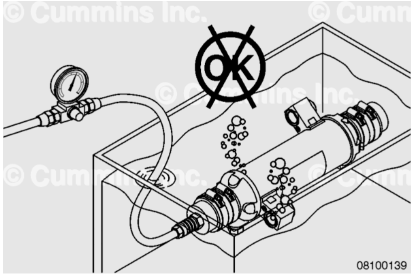
Inspect for air leaks at the braze joints of the end caps and any bubbles from the gear oil ports. If leaks are detected, replace the gear oil cooler.

Remove the cooler from the tank.

Shut the air supply OFF. Disconnect the air supply.

Remove the test equipment.

Note : Although the rear mount cooler is shown, this applies to both cooler designs.



⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to reduce the possibility of personal injury.

⚠ WARNING ⚠

Some solvents are flammable and toxic. Read the manufacturer's instructions before using.

⚠ WARNING ⚠

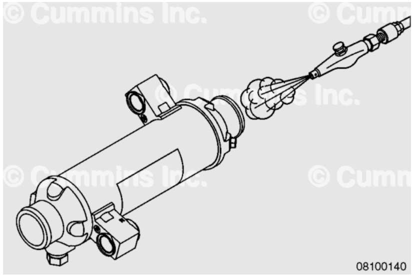
Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

Drain the water from the cooler.

Flush the oil side of the cooler. Use clean solvent.

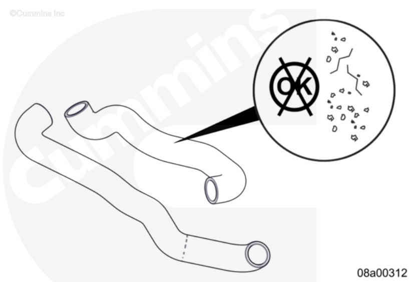
Use compressed air to dry the cooler.

Note : Although the rear mount cooler is shown, this applies to both cooler designs.



Inspect the marine gear cooler sea water hoses for damage. Replace if damage is found.

Inspect the heat shielding installed on the hose near the turbocharger compressor housing and on the aftercooler.



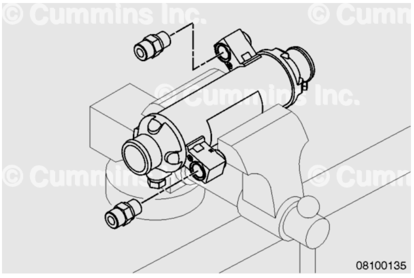
If the shield is torn or sufficiently worn such that the black molded hose is exposed in area to be shielded, the shield should be replaced.

If the shield's outer aluminum film delaminates or significantly degrades, exposing inner white fiberglass sleeve, the shield should be replaced.

Note : If the hose is replaced, a new heat shield should also be installed.

Assemble

Rear Mount Gear Oil Cooler:

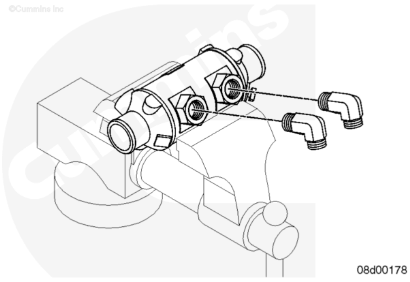


Install the two line fittings into the gear cooler, if removed. Be sure they are oriented in the same direction as they were removed.

Tighten the fittings.

Torque Value: 55 n•m [41 ft-lb]

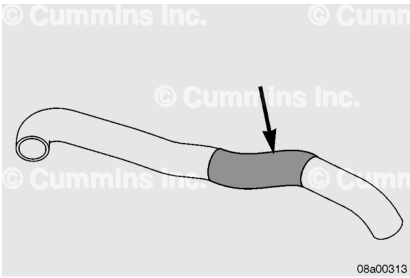
Side Mount Gear Oil Cooler:



If removed, coat the line elbow fitting threads with thread sealant and install into the gear oil cooler. Be sure the fittings are oriented in the same direction as they were removed.

Note : The elbow fittings are pipe thread, therefore there is not a torque value. Once the fitting is hand-tight, turn the fitting another two to three turns to establish sealed threads and proper fitting orientation.

Molded Hose Heat Shield Installation (preferred method):



1. Use a general/multi-purpose spray lubricant that is compatible with rubber. Lubricate the heat exchanger end of the molded hose past the area of hose to be shielded.
2. Insert a long breaker bar into the heat exchanger end of hose to straighten out the hose bend.
3. Slide the heat shield onto the hose from the end straightened with breaker bar.

Note : An alternative method of installing the heat shield is to cut the heat shield, wrap it around the hose, and attach it using metal zip ties. Be sure that the cut

seam of the heat shield is positioned away from the turbocharger compressor housing.

Use the following procedure for aftercooler heat shield installation. Refer to Procedure 010-005 in Section 10. (/qs3/pubsys2/xml/en/procedures/40/40-010-005.html)

Install

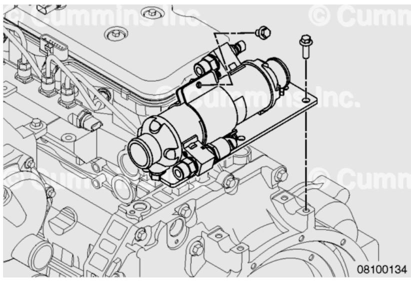
Rear Mount Gear Oil Cooler:

Install the cooler bracket-to-flywheel housing mounting capscrews. Tighten the capscrews.

Torque Value: 44 n•m [32 in-lb]

Install the marine gear oil cooler. Install the bracket-to-cooler mounting capscrews. Tighten the capscrews.

Torque Value: 44 n•m [32 ft-lb]

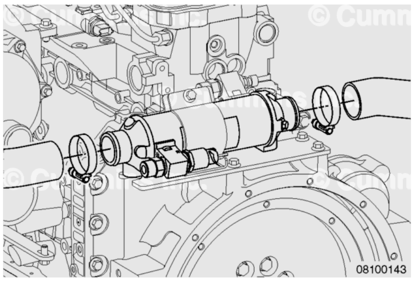


Install the marine gear oil cooler sea water discharge hose. Tighten the hose clamp.

Torque Value: 8 n•m [71 in-lb]

Install the aftercooler discharge hose. Tighten the hose clamp.

Torque Value: 8 n•m [71 in-lb]

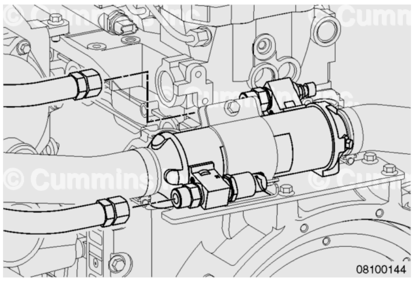


Install the transmission oil supply and return hoses to the gear cooler.

Torque Value: 30 n•m [22 ft-lb]

Install the sea water supply for the shaft log seal to gear oil cooler, if equipped.

Connect the gear oil pressure and temperature sensor wiring harness connector, if equipped.

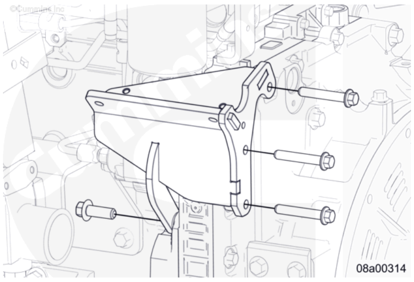


Side Mount Gear Oil Cooler:

If removed, install the gear cooler mounting brackets and capscrews.

Install the bracket capscrews at the flywheel housing.

Torque Value: 46 n•m [34 ft-lb]



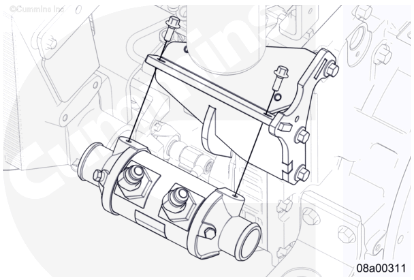
Install the bracket capscrew at the rear gear housing accessory cover.

Torque Value: 62 n•m [46 ft-lb]

Install the wiring harness clips to the gear oil cooler mounting bracket.

Install the marine gear oil cooler and capscrews to the mounting bracket.

Torque Value: 23 n•m [204 in-lb]



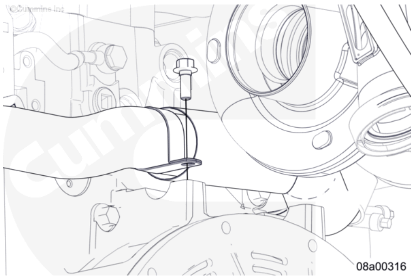
Install the gear oil cooler sea water hoses. Tighten the hose clamps.

Torque Value: 8 n•m [71 in-lb]

Install the hose support clip at the flywheel housing. Tighten the capscrew.

Torque Value: 30 n•m [22 ft-lb]

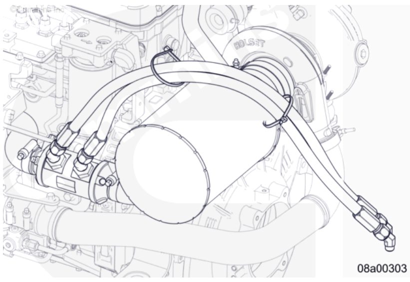
Note : The heat shield is located to the right of the hose support clip and below the turbocharger compressor housing.



Install the transmission oil supply and return lines to the gear oil cooler fittings. Route the lines over the air filter as shown.

Torque Value: 40 n•m [30 ft-lb]

Install the sea water supply for the shaft log seal to gear oil cooler, if equipped. Connect the gear oil pressure/temperature sensor wiring harness connector, if equipped.



Finishing Steps

⚠ WARNING ⚠

Batteries can emit explosive gases. To reduce the possibility of personal injury, always ventilate the compartment before servicing the batteries. To reduce the possibility of arcing, remove the negative (-) battery cable first and attach the negative (-) battery cable last.

- Install the air cleaner.
- Connect the batteries. See equipment manufacturer service information.
- Open the sea water inlet valve(s).
- Operate the engine. Check for leaks.

Last Modified: 09-Mar-2017